

INDUSTRIAL TRAINING REPORT

A TRAINING REPORT

Submitted by

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(24BEC10015)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION



CHANDIGARH UNIVERSITY

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BONAFIDE CERTIFICATE

Certified that this training report **INDUSTRIAL TRAINING** is the bonafide work of DEVANSHI (24BEC10015) who carried out the training under WONDER AUTOMATION, MOHALI.

Signature
Internal Examiner

Signature
External Examiner

Signature
HOD

ACKNOWLEDGEMENT

I would like to thank Wonder Automation, Mohali for providing me the opportunity to undertake my Industrial Training with them. I have gained a wider knowledge of Industrial Automation and how the use of modern technology applies to Industrial Automation and Manufacturing Industry through this internship.

During the training period, I had exposure of PLC Programming, Ladder Logic Design, Industrial Integration and Hands-on Programming. I learned to apply theories to practice in this practical learning.

I would like to thank Pooja Shatriya, my training mentor at Wonder Automation for constant guidance, encouragement and feedback during the Internship. I am also thankful to the technical and administration members of Wonder Automation for their co-operation and conducive working atmosphere.

I would like to extend my heartfelt thanks to Chandigarh University and the Department of Electronics and Communication Engineering for giving me the opportunity to utilize my classroom learning in the industry and to my faculty members and training coordinators for their guidance.

Finally, I recognize my colleagues for their co-operation and fellowship. This internship will provide me the skills and experience that will help me in my future career in the field of Industrial Automation and Control Systems.

ABSTRACT

The industrial training that was done at Wonder Automation, Mohali gave an in-hand exposure to the field of Industrial Automation. The main purpose of this training was to turn into practice the knowledge that has been acquired in academic studies.

The training course covered the important areas of PLC Programming, Ladder logic Design and Industrial Integration. The following activities were given special focus: Siemens PLCs, industrial automation methodologies, hardware configuration, programming, system integration and troubleshooting techniques. Practical sessions and live demonstrations helped in understanding the functioning of industrial automation systems.

This learning experience enabled me to understand the process of industrial automation and application of PLC based control system in real-time applications. The programme also helped in developing professional skills like analytical reasoning, team work and problem solving.

To conclude: The exposure to industry at Wonder Automation, Mohali has helped strengthen the relation between academics and industry, and has laid a strong foundation for me to pursue my career in the field of Industrial Automation and Control Systems.

CERTIFICATE

SR.NO: WATD/11/7/3605

Wonder
AUTOMATION

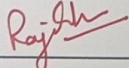
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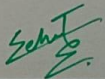
THIS CERTIFICATE IS AWARDED TO

DEVANSHI

for Successfully Completing
4/6 Weeks Specialised Training in Industrial Automation covering
PLC Programming

From 02-06-2026 To 07-07-2026

 
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1. INTRODUCTION

Industrial training is an essential part of engineering education and gives students a chance to understand how theory can be put to use in a real industrial setting. It is a link between theory and practice by introducing students to the latest technologies, industrial process and engineering standards. This type of training provides not only with technical skills but also with the analytical thinking and problem-solving skills, as well as confidence needed for the fast-changing industrial world.

I did my 6 week industrial training at Wonder Automation which is one of the leading Industrial Automation, PLC Programming, System Integration, and Motion Control Company. The training was designed to give practical knowledge about automation systems as well as to introduce to the students Indian Industrial practices through hands-on and live demonstration method.

With Industry 4.0 in the picture, automation has become the key to efficient, intelligent and reliable industrial operations. Programmable Logic Controller (PLC), Human Machine Interface (HMI), Supervisory Control and Data Acquisition (SCADA), industrial sensor and control systems are changing the landscape of manufacturing and process industry. With the push of industries to become more productive, quality and safe in usage of automated systems, engineers with hands-on experience in these technologies are in great demand. This training was insightful and informative about these developments and their use in practice.

It was a good training for me to complement with my academic knowledge on topics like Digital Electronics, Control Systems, Embedded Systems etc. and also improved my problem solving skills, Teamworking skills, Communication skills and Technical skills. This report is a comprehensive overview of the training programme including Company profile, Objectives, Methodology, Technical learning, Practical learning, Outcomes and Overall learning during internship..

2. COMPANY PROFILE

2.1 About the Organization

Wonder Automation, Mohali, Punjab is one of the premier companies offering Industrial Automation, PLC Programming, System Integration, Motion Control and Industrial Training services. In association with Wonder Systems India Pvt. Ltd., the company has more than 20 years of automation solutions and industry oriented technical training experience. It is committed to the application of actual automation solutions, as well as developing skilled personnel through hands-on experience.

2.2 Areas of Expertise and Services

- PLC Programming and Industrial Automation
- Ability to develop a system integration and control panel.Skill in developing a system integration and control panel.
- Developing HMI and SCADA applications
- Determine the motion control and drive systems required for a specific application.
- This course will help students understand the basics of industrial commissioning and troubleshooting.

2.3 Vision, Mission and Core Values

Vision: To deliver quality automation solutions and industry relevant practical training

Mission: To build highly skilled automation professionals and provide innovative solutions to make itself a trusted leader in the industrial automation sector.

Core values: Innovation, Quality, Integrity, Teamwork, Continuous Learning.

2.4 Industries Served

Wonder Automation is used for a range of industries, such as:

- Manufacturing Industries
- Process Industries
- Food Processing
- Pharmaceutical Industries
- Packaging Industries
- Educational Institutions

Its automation solutions come in the aim of efficiency, productivity, process reliability in various industrial environments.

2.5 Tools, Platforms and Technologies Used

The company has sophisticated automation hardware and software to design and deploy industrial control systems.

Hardware: Siemens PLCs, HMI, VFDs, Sensors, Relays, Control Panels

Software: Siemens TIA Portal, PLC Programming Software, HMI Software, SCADA

Technologies: PLC programming, Ladder Logic, Industrial automation, Motion control, System integration and Fault finding.

2.6 Training and Skill Development Activities

A Wonder Automation is offering the engineering students and professionals Industrial training and Internship training programmes. PLC Programming, Ladder Logic Design, HMI, SCADA, Industrial Integration and fault finding using Siemens PLCs and live industrial setups are covered in the training. The programme is a mixture of theory and practical, enabling the trainees to build up their industrial experience and technical skills.

3. OBJECTIVE OF THE TRAINING

The main aim of the 6-weeks industrial training was to do a gap finding between theory learning in academic and industrial application at Wonder automation, Mohali. The main objectives of the training were to build up their technical skills, to reinforce their understanding of industrial automation and to prepare students for a professional life in the automation sector. The specific objectives as follows:

1. To understand the importance of the Industrial Automation systems

- To gain knowledge of the working and structure of a PLC based automation system..
- Watch how automation technologies are applied in actual industrial production.

2. To develop an understanding of the theory behind PLCs

- Write Ladder Logic programs with Siemens PLCs.
- Run and troubleshoot automation programs in real-world situations.

3. To grasp the concept of industrial hardware and techniques related to system integration

- Understand the role of PLCs, HMI, sensors, relays, and VFDs.
- To understand the integration of different hardware components in an Automation system.

4. To further develop troubleshooting and problem solving skills

- Identify and rectify common faults in PLC programs and wiring.
- Develop a systematic approach to industrial fault diagnosis.

5. To get to know industrial software tools.

- Program a PLC using Siemens TIA Portal.
- Using Kinco Builder design and configure HMI screens..

6. To increase awareness about industrial safety practices.

- Work with automation equipment safely in accordance with electrical safety rules.
- To have an awareness of safe working and professional working practices..

7. To acquire knowledge about specific work and work-related situations

- Learn how to better work cooperatively, communicate and manage time.
- Build confidence through practical learning and technical discussions..

8. To foster a spirit of collaboration among students, faculty, and industry partners

- Use Digital Electronics and Control Systems concepts in real automation industry applications.
- Recognize the application of the theoretical knowledge to solve the practical problems in engineering..

4. TRAINING METHODOLOGY

4.1 Classroom Lectures and Conceptual Sessions

The training began with classroom training sessions by experienced trainers, giving a good theoretical base on industrial automation. These sessions were designed to cover the fundamentals and provide an understanding of the principles of an industrial control system, before then working through the practicality of implementing it.

Topics covered included:

- Basics of Industrial Automation and PLCs
- Ladder Logic Programming
- Industrial Safety Practices

Outcome: Conceptual sessions enhanced the knowledge of automation principles and further prepared the trainees for programming and hardware implementation tasks.

4.2 Practical Demonstrations and Hands-on Training

Trainees were then provided with practical sessions involving Siemens PLC and industrial automation equipment and given the chance to understand the concepts through hands on experience. Trainers showed the trainees how to program, wire and operate the system before letting the trainees perform these tasks.

Key practical components included:

- PLC wiring and hardware connections
- Ladder Logic Programming
- Input and Output testing
- Fault finding and troubleshooting

Outcome: This Hands-on training had helped to enhance the level of expertise and confidence in operating industrial automation equipment.

4.3 Project-Based Learning Approach

The training comprised of the practical assignments and small automation tasks which involved the trainees to apply the concept learned in the training session. These activities stimulated logical thinking, systematic problem solving and teamwork in developing automation solutions.

Activities included:

- Designing a program to control automation using PLCs
- Develop and test control logic
- The practical implementation of the automation exercises.

Outcome: Project-based learning facilitated logical thinking, accuracy in programming and practical problem solving.

4.4 Simulation and Software Training

Software training was performed with the help of Kinco's PLC programming development software - Kinco Builder. The PLC programs were designed, tested in software and verified on hardware prior to implementation. The HMI screens were designed using Kinco Builder to display and control industrial processes and achieve a good interaction between operators and automation systems.

Outcome: Software training improved programming efficiency, HMI development skills and the design of reliable industrial automation systems.

4.5 Guided Mentorship and Supervised Practice

All ongoing training was supported with constant guidance from experienced trainers both in theory and practical sessions. They frequently checked the assignments, shared technical questions and offered suggestions for improvement to deepen comprehension and technical ability.

Mentorship included:

- One-to-one guidance in Practice sessions
- Feedback on programming tasks
- Troubleshooting of errors

Outcome: Ongoing mentorship helped to speed learning, build technical confidence and foster a systematic engineering approach.

4.6 Industrial Exposure and Observation

Finally, actual demonstration of industrial automation systems and its applications in manufacturing industries was made and discussed. The trainees witnessed the integration of PLC, HMI, sensors, and industrial control panels in the automation of industrial processes.

Outcome: Industrial exposure gave an insight of the latest concept of automation and gave an excellent linkage between academic knowledge and industrial application.

5. WEEKLY TRAINING SUMMARY

The six-week Industrial Training was designed to be progressive with the introduction of the basics of Programmable Logic Controllers (PLC) and slowly moving into the topic of industrial programming, automation logic, and applications. The training also included practical sessions focusing on the use of the Kinco Builder software, allowing the trainees to apply the concepts they had learned in a hands-on manner.

Week 1: Introduction to PLC and Programming Fundamentals

Objective

The goal of the initial week was to establish a solid foundation in the field of industrial automation, and to introduce the basic concepts of Programmable Logic Controllers (PLCs). The week emphasized introducing the PLC architecture, various programming languages, and basic ladder logic, the foundation of industrial PLCs.

Content Covered

- Introduction to Programmable Logic Controller (PLC) and its application in industry.
- Understand the parts and operation of a PLC, including the CPU, input/output modules, memory and power supply.
- Understanding of types of PLC programming languages as per IEC 61131-3 with emphasis on ladder diagram (LD).
- Basics of Ladder Diagram programming: rails, rungs, contacts, coils, program execution.
- Learning about AND, OR, NOT, NAND, NOR, XOR and XNOR logic gates, and how they can be implemented in ladder logic.
- Simple Ladder logic program design using logic gates practical exercises.
- Introduction to the Kinco Builder Software interface and development of simple PLC programs.

Outcome

After a week, I got an idea about the basics of PLCs and the use of PLCs in industrial automation. I got to know the Kinco Builder software, and made and simulated basic ladder logic programs with various logic gates.

Week 2: PLC Logic Design and Control Operations

Objective

The second week was used to build up logical programming skills needed for industrial automation. The goal was to learn about the use of interlocking, sequential operations, memory, and switching logic for PLC control of machines.

Content Covered

- Understanding of Interlocking for safe operation of industrial equipment – concept and implementation.
- Use sequence programming techniques to control multiple operations in a specified sequence.

- Storing of intermediate logic conditions and program states using Memory Bits.
- PLC toggle switch logic ON/OFF Control.
- Learn how to make ladder logic programs with more than one input and output.
- Hands-on practice in the design of sequence control and interlocking circuits.
- Programming and simulation with the Kinco Builder software.

Outcome

After two weeks, I could create ladder logic programs with sequential operations, interlocking conditions, memory bits and toggle switch functionality. This deepened my understanding on industrial machine control and logical programming..

Week 3: Timers, Counters, and Industrial Applications

Objective

The goal of the 3rd week was to present of the time-based and event-based control operations that are often encountered in industrial automation systems. The focus of training was on the practical aspects of timers, counters and their use in real world control applications.

Content Covered

- Introduction of PLC Timers and types of timers.
- Programming and use of ON-delay and OFF-delay timer instructions.
- Introduction to PLC Counters and counting operations.
- Programming Up Counters and Reset Operations.
- Construction of a Traffic Light Control System using Ladder logic.
- Testing and simulation of timer/counter programs for automation.
- Application of the same with Kinco Builder software..

Outcome

At the end of this week, I got hands-on experience to use timer and counter based automation systems. I have been able to design and test the traffic light control program successfully and learned more about time-controlled industrial processes.

Week 4: Function Block Programming and Industrial Automation

Objective

The aim of the fourth week was to learn the advanced programming methods of PLC using Function Block Programming (FBD) and also apply all the previous skills learned in practical industrial automation applications. The emphasis was on more efficient programming, system development, and troubleshooting abilities.

Content Covered

- Introduction to Function Block Diagram (FBD) programming.
- Comparison between Ladder Diagram (LD) and Function Block Diagram (FBD).
- Developing automation logic by means of function blocks.
- Incorporation of timers, counters, memory bits and interlocking logic in the function block programming.
- Program testing, debugging and troubleshooting procedures.
- Real world application and simulation of full PLC automation solutions.
- Working with the Kinco Builder software to design, test and verify PLC programs..

Outcome

I learned how to create, test and troubleshoot PLC programs independently using Ladder Diagram Programming and Function Block Programming at the end of the training. Developed confidence in the use of PLC programming concepts to solve industrial automation problems and improved hands-on capabilities with the Kinco Builder software.

Week 5: SCADA Systems**Objective:**

To understand the basic concepts of SCADA (Supervisory Control and Data Acquisition) and learn how it is used for monitoring and controlling industrial processes.

Content Covered:

- Introduction to SCADA and its role in Industrial Automation.
- Simple architecture of SCADA and its key components.
- Inter-PLC/Inter-SCADA communications.
- Developing process monitoring graphical screens.
- Real time parameter and machine status display.
- Knowing how to use alarms, indicators and basic data monitoring.

Outcome:

Learned basics of SCADA systems and how processes in industry can be monitored and controlled via graphical interface.

WEEK 5 PROJECT EXTENSION

Design and Validation of a Web-Based PLC Traffic-Light Simulator

Supplementary learning project - developed after the July 2026 training

This chapter documents Signal PLC Studio, a browser-based educational simulator that demonstrates a timer-driven two-road traffic-light sequence. It extends the concepts covered during PLC and SCADA training by presenting a visual intersection, a ladder-style output illustration, and a simulated address map in one interface.

The simulator was developed after the formal training period. It is not evidence of a project completed at the training company, does not execute an IEC 61131-3 PLC program, and is not connected to a PLC or road equipment. The report keeps this distinction explicit so that the software result is not confused with a commissioned industrial installation.

The work applies sequence design, input/output mapping, interlocking concepts, interface design, and software testing. The following pages describe the requirements, phase model, implementation, observed tests, limitations, and opportunities for future work.

Project item	Description
Project	Signal PLC Studio - browser-based traffic-light training simulator
Primary purpose	Demonstrate a four-phase timed sequence and PLC-oriented views
Implementation	React application; timer and state execute in the browser
Evidence	Live simulator screenshot, diagrams, and repeatable UI checks
Safety boundary	Educational display only; no real PLC or field devices

2. Problem Definition and Design Requirements

Traffic-light control is a familiar example of sequential automation: outputs must change in a defined order, remain active for controlled durations, and prevent incompatible directions from receiving a green indication at the same time. The simulator turns this abstract control problem into a visible training exercise that can be observed without wiring a physical controller.

The design was scoped to a simplified two-approach intersection. It intentionally excludes vehicle detection, pedestrian demand, adaptive timing, emergency-vehicle pre-emption, cabinet monitoring, and jurisdiction-specific signal standards. These exclusions keep the demonstration focused on basic sequence and interlock concepts.

Requirement	Design response
Show both traffic directions	Intersection view shows North/South and East/West lamps.
Sequence timed indications	Four phases use 8-second green and 3-second amber intervals.
Avoid opposing greens	The phase model assigns green to one approach while the other is red.
Expose control state	The interface presents phase, timer, cycle timeline, ladder view, and I/O map.
Allow safe demonstration	Run, pause, and reset controls operate only the software model.
Make limitations visible	The project is labelled simulated and educational, not road-use equipment.

These requirements are functional acceptance points, not a safety certification. A real traffic controller requires requirements traceability, jurisdictional design review, fail-safe hardware, independent verification, environmental qualification, and formal commissioning.

3. Traffic-Phase Sequence

The cycle uses four mutually exclusive states. North/South receives the first green interval; its amber interval provides a transition before East/West green. East/West then receives its own amber interval before the cycle returns to the first state. Red is shown for the opposing approach throughout each green and amber phase.

Phase	North / South	East / West	Duration
1	Green	Red	8 s
2	Amber	Red	3 s
3	Red	Green	8 s
4	Red	Amber	3 s

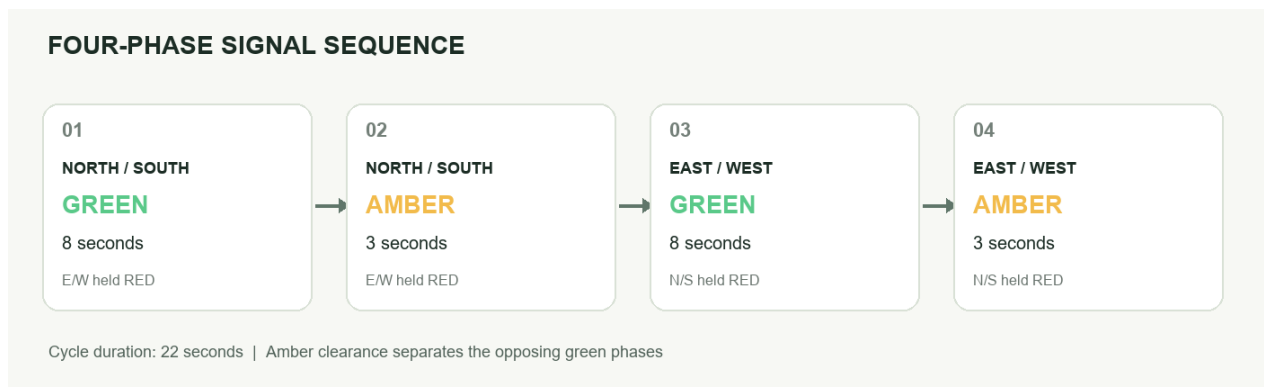


Figure 1. Four-state sequence and nominal phase durations.

The nominal cycle totals 22 seconds: two green intervals of 8 seconds each and two amber intervals of 3 seconds each. The displayed duration is a teaching parameter for the simulator. It is not a timing recommendation for public roads and does not include engineering allowances for clearance intervals, pedestrian crossings, approach geometry, or local regulations.

4. Timing, Interlocking, and State Transitions

The conceptual controller can be represented as a finite-state sequence. In each state, a timer accumulates elapsed time. When the configured duration expires, control advances to the next state and resets the timer for that phase. A pause control stops the demonstration; reset returns the interface to its initial standby condition.

The key interlocking invariant is that the North/South and East/West green outputs must never be active together. In this simplified sequence, phase exclusivity enforces the invariant by construction: only one phase bit is selected at a time, and each green output is derived from its corresponding green phase. The opposing red indication is derived for both states of the active approach.

Transition	Expected control action
Phase 1 -> 2	N/S changes from green to amber; E/W remains red.
Phase 2 -> 3	N/S changes to red before E/W green is displayed.
Phase 3 -> 4	E/W changes from green to amber; N/S remains red.
Phase 4 -> 1	E/W changes to red before N/S green is displayed.
Reset	Return to the initial paused/standby state and default phase.

A production controller would also define behavior for invalid state, timer fault, power interruption, output mismatch, restart, and emergency stop. This browser demonstration does not implement or validate those hardware fault responses.

5. Simulated Input and Output Address Map

The address panel uses PLC-style labels to connect the visible sequence with common discrete I/O terminology. The labels are a software representation only: the application does not communicate with a PLC rack, field wiring, or lamp drivers.

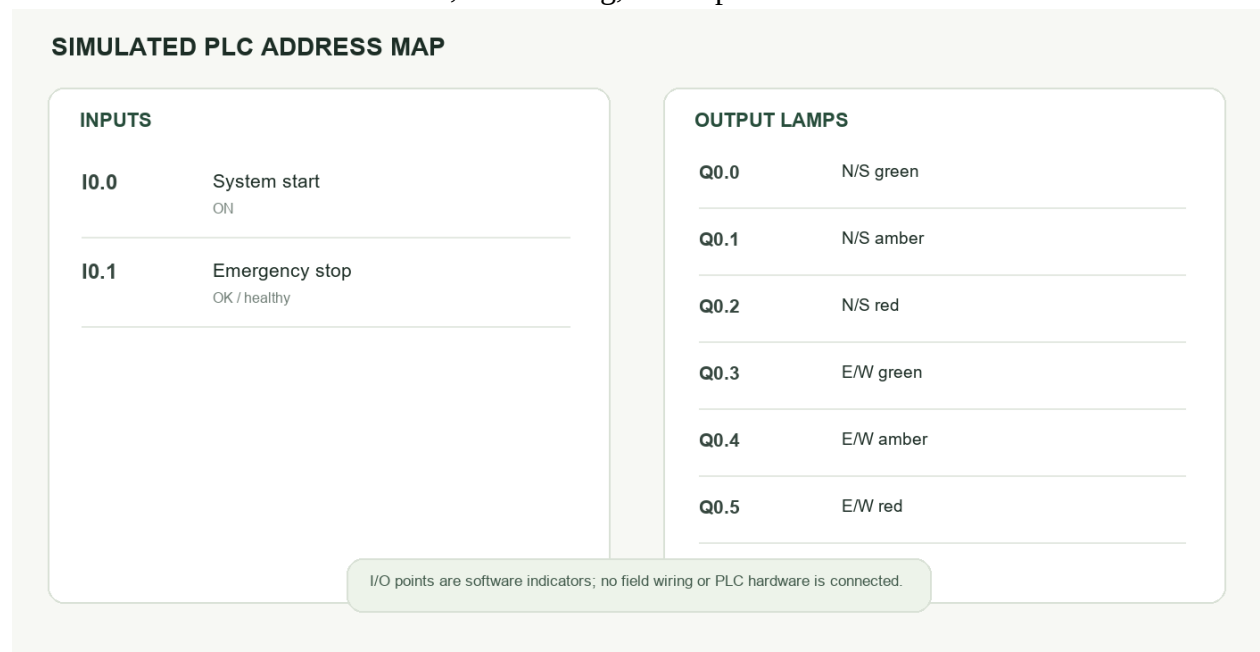


Figure 2. Address labels presented by the simulator; states are software indicators.

Address	Point	Type / meaning
I0.0	System start	Simulated digital input
I0.1	Emergency stop	Displayed healthy; no physical safety circuit
Q0.0-Q0.2	N/S green, amber, red	Three simulated lamp outputs
Q0.3-Q0.5	E/W green, amber, red	Three simulated lamp outputs
M0.0-M0.3	Phase 1-4	Internal phase markers used in the ladder illustration

During operation, the I/O view derives lamp states from the active software phase. I0.1 is shown as healthy in the demonstration; an actual normally-closed emergency-stop circuit must be designed, wired, and validated independently.

6. Ladder-Style Control Representation

A ladder-oriented view helps relate sequence states to output coils. The illustration maps each phase marker to one or more lamp labels: M0.0 selects North/South green and East/West red; M0.1 selects North/South amber and East/West red; M0.2 selects East/West green and North/South red; M0.3 selects East/West amber and North/South red.

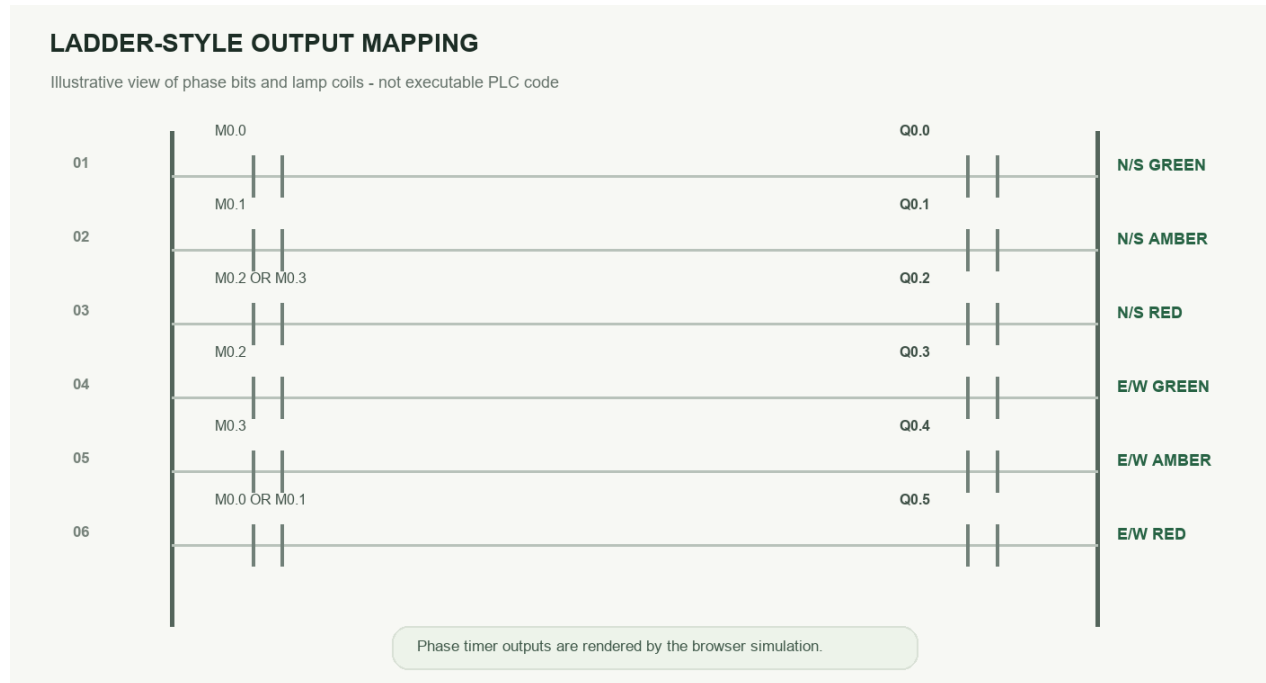


Figure 3. Simplified ladder-style mapping shown for instruction; it is not a PLC program.

The diagram communicates the relationship between memory bits and lamp outputs, but it omits controller-specific timer instructions, scan semantics, retentive behavior, safety-rated design, and downloadable PLC syntax. The website renders this view from application state. It should therefore be described as a ladder-style visualization rather than executable ladder logic.

The separation between the diagram and the running model is important: a resemblance to a programming language does not make the drawing validated, portable, or safe to deploy.

7. Software Architecture and State Flow

Signal PLC Studio is a client-side React application. Operator actions update application state; a browser timer advances the active phase while the simulation is running. The phase state and elapsed time then drive the intersection lamps, countdown, phase timeline, I/O indicators, and active-rung highlighting.

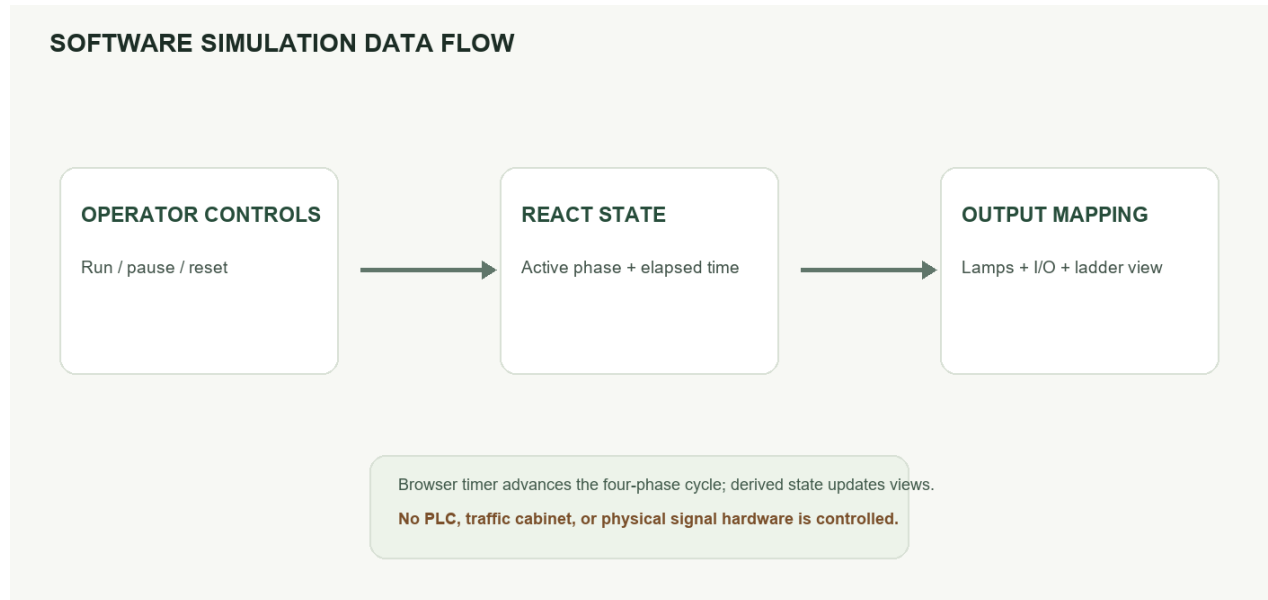


Figure 4. High-level browser simulation data flow.

This architecture supports a fast, self-contained demonstration and avoids the setup needed for a physical controller. It also establishes a strict boundary: browser timing is not real-time control, and a UI state cannot guarantee output behavior under operating-system delay, tab suspension, device failure, or power loss.

The codebase is built with React and Vite. A GitHub Actions workflow builds the static site and publishes it through GitHub Pages. The hosted application is useful for viewing the interface, while source files remain available in the project repository.

8. Operator Interface and Demonstration Procedure

The main screen combines the active phase, remaining time, cycle duration, a live intersection illustration, a phase timeline, and a ladder preview. Navigation provides dedicated ladder-logic and I/O-map views. Run, pause, and reset controls make it possible to repeat the sequence during a demonstration.

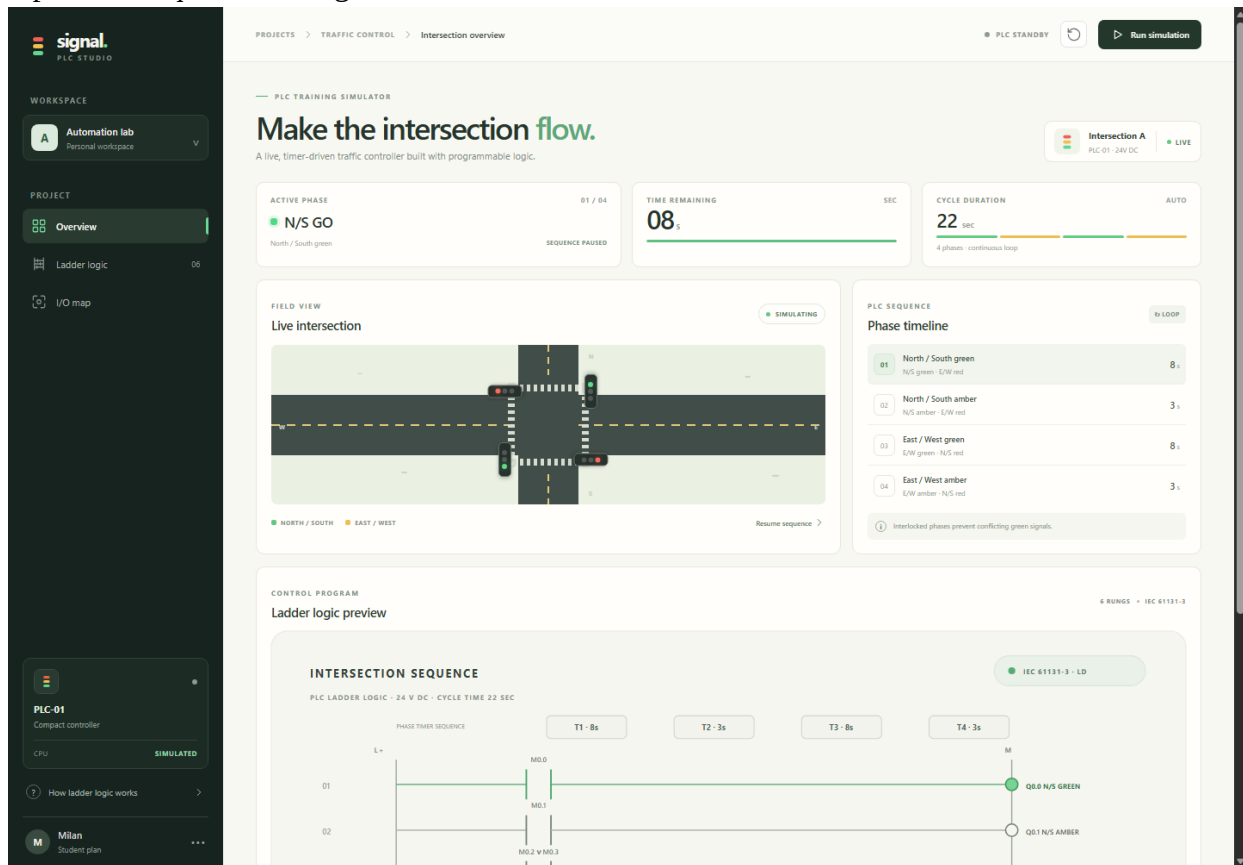


Figure 5. Captured overview of the hosted Signal PLC Studio simulator.

A typical demonstration starts from standby, selects Run, and observes the North/South green-to-amber transition followed by the East/West phases. Pause freezes the instructional display; Reset returns it to the initial condition. The operator can then inspect the lamp-address map and compare active outputs with the phase timeline.

The screenshot is genuine output from the deployed simulator. The circuit and address diagrams elsewhere in this chapter are explanatory figures and should not be interpreted as photographs of PLC hardware.

9. Verification Approach and Test Results

Verification focused on observable user-interface behavior. The checks were performed against the browser simulator and its deployed interface; they do not constitute hardware-in-the-loop testing, PLC validation, or road-signal commissioning.

Check	Procedure	Observed result
Start and cycle	Run the sequence and observe phase changes.	Four phases advance in the defined order.
Timing display	Observe green and amber countdowns.	Nominal 8 s green / 3 s amber durations are shown.
Pause	Pause during an active phase.	Display remains in the selected phase until resumed.
Reset	Reset after starting the sequence.	Simulation returns to standby/default state.
I/O mapping	Compare lamp state with the address panel.	Displayed outputs correspond to the selected phase.
Image export	Export the circuit diagram as PNG.	PNG download completes in the browser.

The test results confirm that the listed interface behaviors worked during the project checks. They do not establish timing accuracy under real-time constraints, electrical compatibility, safety integrity, or correct behavior of a physical controller.

10. Results and Discussion

The finished application provides a repeatable visual demonstration of the selected traffic-control sequence. The overview shows which direction has permission to proceed, the countdown indicates the programmed duration, and the timeline makes the next phase visible. The ladder-style and I/O views offer alternate ways to interpret the same simulated state.

Active phase	Expected active lamps	Opposing approach
N/S green	Q0.0 N/S green; Q0.5 E/W red	Red
N/S amber	Q0.1 N/S amber; Q0.5 E/W red	Red
E/W green	Q0.3 E/W green; Q0.2 N/S red	Red
E/W amber	Q0.4 E/W amber; Q0.2 N/S red	Red

At the selected timings, the nominal cycle is 22 seconds, of which 16 seconds are green intervals and 6 seconds are amber intervals. This arithmetic describes only the simulator configuration; real timing plans depend on traffic engineering and applicable standards. The result demonstrates how an automation concept can be decomposed into states, transitions, outputs, and operator feedback. It also provides a visual artefact that can be shared for review without claiming a physical installation.

11. Limitations and Safety Considerations

This application is an educational software simulation, not a safety-related controller. It must not be connected to or used to operate public-road signals, private traffic-control equipment, or any device whose failure could cause injury or property damage.

Limitation	Implication
No PLC hardware	No electrical outputs, input wiring, or controller scan are tested.
Browser timer	Scheduling may vary with load, tab visibility, or operating system.
Simplified junction	No vehicle/pedestrian detectors or adaptive traffic logic.
Illustrative ladder	The diagram is not compiled or validated IEC 61131-3 code.
No formal safety case	No hazard analysis, independent verification, or certification.
No field commissioning	No cabinet, signal heads, communications, or site work.

A real deployment would require qualified traffic-control and functional-safety engineers, an approved requirements specification, an appropriate certified controller, independent conflict monitoring, fail-safe design, physical inspection, and jurisdictional approval. The simulator intentionally has no interface for connecting to field equipment.

12. Skills Gained, Challenges, and Future Enhancements

The project reinforced the connection between sequence-control concepts and operator-facing visualization. Converting a written sequence into explicit phases helped make timing and interlocking assumptions testable. Mapping the same state into a field view, ladder-style diagram, and I/O table showed how a control concept can be explained to different audiences.

Learning area	Application in this project
Sequential logic	Defined four ordered phases and phase transitions.
Timers	Applied separate green and amber durations.
I/O concepts	Associated phase state with simulated lamp addresses.
Testing	Checked run, pause, reset, phase order, I/O state, and export.
Technical communication	Documented scope, evidence, and safety boundary.

The main design challenge was keeping the displays consistent as the active phase changed. Deriving the lamps and indicators from one application state reduces duplicated logic in the interface. Another challenge was making the ladder view informative without overstating its status as an actual PLC program.

Future educational improvements could add pedestrian phases, detector inputs, configurable timing, event logs, accessibility controls, and a fault-injection mode. Any hardware-oriented extension would be a separate engineering project with its own risk assessment and validation plan.

13. Conclusion and Project References

Signal PLC Studio translates a basic two-road traffic-light sequence into an interactive browser demonstration. Its four phases, timer display, simulated I/O map, ladder-style visualization, and downloadable circuit image make introductory automation concepts easier to observe and discuss.

The documented checks support the stated software behavior, while the limitations section prevents those results from being mistaken for industrial validation. The project was created as supplementary work after the July 2026 training and does not replace or revise the six-week training activities described elsewhere in this report.

Reference	Resource
Project repository	https://github.com/thedevanshi/signal-plc-studio
Hosted simulator	https://thedevanshi.github.io/signal-plc-studio/
Project technology	React and Vite; deployment through GitHub Actions / GitHub Pages
Application purpose	Educational simulation only; not for real traffic control

The overall outcome is a working, shareable software artefact that connects timer-based sequence logic with an operator interface. Its value is instructional: it demonstrates concepts, supports discussion, and provides a foundation for further controlled experimentation.

Week 6: Implementation of Automation Project

Objective:

To transfer the knowledge gained from the training to create and execute a realistic project for industry automation.

Content Covered:

- Selection and knowledge of requirements for the automation project.
- Writing the control code with PLC programming.
- Creating the ladder logic for the desired sequence of operation.
- Assigning the roles of inputs and outputs, and testing the program.
- Installing the PLC to the desired field devices.
- Testing, troubleshooting and making required changes.
- Showing the last working of the automation project.

Outcome:

Successfully used PLC programming and automation concepts for a real-life project. Gain knowledge of control logic and troubleshooting, input-output interfacing and implementation of an industrial automation system.

6. CHALLENGES FACED AND SOLUTIONS

1. Understanding PLC Programming Logic

Challenge:

Many concepts of PLC programming like ladder logic, contacts, coils, timers, counters, and addressing memory were not easy to comprehend at the start of the industrial training. Programming PLCs was new to me, and I was not yet familiar with how a process works or how various instructions interact to perform the process. This had a slight effect on the speed of my ladder logic program development and debugging.

Solution:

The solution was to continually make changes to the theoretical concepts and practice writing simple ladder logic programs on a regular basis. The trainers taught the concepts through practical examples and explained the functioning of each instruction. By practicing and testing small programs over and over, I got more confident and proficient in programming.

2. Troubleshooting Wiring and Hardware Connections

Challenge:

In practical sessions, I encountered problems with making the correct wiring connections between the modules of the PLC and the devices such as sensors, switches, relays and indicator lamps. Faulty wiring could lead to no communication or unintended output, making it hard to pinpoint the problem.

Solution:

It was explained to us that we need to be careful in reading the electrical wiring diagrams and check all connections before we turn on the system. Learned to test for terminal numbers, polarity of the power supply and continuity properly. This systematic approach greatly minimized wiring errors and my troubleshooting skills.

3. Learning HMI Development Using Kinco Builder Software

Challenge:

The development of the HMI screens using Kinco Builder software was not an easy task because I didn't have experience with industrial visualization software. Setting up the communication parameters and assigning PLC addresses and creating user-friendly screens demanded an understanding and level of attention.

Solution:

Working with the trainers, I was able to learn the software from the ground up, starting with developing simple screens and then progressing to buttons, indicators, alarms, and data displays. With practice I got a clear idea of what properties objects had, how to communicate with them and how to download projects. After that, I got comfortable in developing simple and functional HMI interface.

4. PLC–HMI Communication and Integration Issues

Challenge:

Occasionally communication errors arose when setting up the communication between the PLC and the HMI, or between the HMI and the wrong address in the PLC, or when setting up serial communication parameters incorrectly. This caused the HMI not to display and control the PLC variables correctly.

Solution:

The trainers showed a step by step process to analyze the communication problems by checking baud rate, COM port setting, slave ID, communication protocol and PLC memory addresses. By following this troubleshooting procedure some configuration problems were identified promptly, and communication between the HMI and the PLC was successful.

5. Time Management During Practical Assignments

Challenge:

The theoretical training coupled with the tasks was at times challenging to balance in the allocated training time. To program the PLC, wire the hardware, test it, as well as complete the documentation on the same day, was a challenge in some of the more complicated exercises, and it was a positive learning experience to plan for efficient completion.

Solution:

I enhanced the techniques of time management, by designing the program logic in advance, and keeping notes organized for future use. Breaking tasks down into smaller steps and a planned sequence enabled me to complete tasks more efficiently. This helped me be more efficient, and avoid wasting valuable time.

6. Debugging PLC Programs

Challenge:

In the practical application, some timers did not get the output due to logical errors like an incorrect timer setting, wrong choice of contacts and improper addressing. It took time and careful analysis to determine the exact cause of these errors.

Solution:

To debug programs systematically, I learned to observe ladder logic online, to check the status of I/O points, and to conduct individual testing of each part of the programs. Logical thinking was encouraged by the trainers rather than trial and error. Through the iterative debugging process, I learned to become more analytical and gained confidence in quickly identifying and fixing programming errors.

7. SKILLS ACQUIRED

7.1 Technical Skills

- **PLC Programming** — Learned to write and debug ladder logic programs with timers, counters and I/O instructions for industrial automation applications.
- **Control Panel Wiring** — Completed hands-on training to wire Industrial PLCs, sensors, relays, contactors, switches and indicator lamps as per Industrial standards.
- **HMI Development** — Designed and configured HMI screens for monitoring and controlling PLC based systems with Kinco HMI.
- **Industrial Devices** — Comprehended how sensors, actuators, relays, and other field devices function and were used in automation systems.

7.2 Software and Simulation Skills

- **Kinco Builder Software** — Designed HMI screens using Kinco Builder Software and tested the interaction between the PLC and HMI.
- **PLC Programming Software** — Developed, tested, monitored and debugged ladder logic programs prior to implementation.

7.3 Analytical and Problem-Solving Skills

I acquired a lot of analytical skills during the training, locating wiring problems, diagnosing ladder logic programs, and communicating between a PLC and an HMI. The trainers explained systematic troubleshooting techniques which helped me to break down the problems logically, find the root causes efficiently and take corrective action as opposed to a trial-and-error approach.

7.4 Industrial and Safety Awareness

Training gave a lot of exposure to industry safety and discipline. I gained knowledge of the need for standard operating procedures (SOPs), electrical safety when dealing with control panels, correct personal protective equipment (PPE) and checking all wiring before energising. These practices gave me an insight into the importance of safety while working with industrial automation systems.

7.5 Communication and Professional Skills

During industrial training, I gained enhanced communication skills and interpersonal skills as I interacted with trainers and fellow trainees on a regular basis, during practical sessions and group discussions. Also learned about the aspect of keeping technical documents in order, preparing technical reports, manners and etiquettes in the workplace, asking pertinent technical questions and working as a team to finish tasks

7.6 Project and Time Management

I enhanced my planning and time management skills as I worked with several practical activities on the training. Developed skills to plan, wire, test, debug and document programming activities in a systematic way, completing training while respecting the training schedule. Practical sessions were more efficient and productive due to prioritizing tasks and having an organized workflow.

8. LEARNING OUTCOMES

1. Technical Learning and Practical Exposure

The training focussed on practical knowledge of industrial automation systems. I had plenty of practical experience in PLC programming, developing the HMI using the Kinco Builder software, industrial wiring, programming sensors and relays, and communicating with both the PLC and the HMI in a real industry, which helped me get a better understanding of how PLC systems work in real industry.

2. Application of Theoretical Knowledge

I learned to use concepts I had studied in Digital Electronics, Microprocessors & Microcontrollers, Industrial Instrumentation and Control Systems in the training. The concepts of theory were made easier to understand when applied to PLC programming, industrial wiring and automation projects.

3. Professional and Soft Skills Development

These skills were a benefit as I worked in a professional environment and became better at communicating, team work, time management and discipline in the working environment

4. Analytical and Problem-Solving Skills

Analytical thinking was reinforced through troubleshooting PLC program, wiring and communication problems. Learnt to analyse problems systematically and use logical solutions rather than trial and error.

5. Project Implementation and Documentation

Creating automation exercises and documenting the practical activities taught me the importance of planning, testing, debugging and keeping proper technical records during a project.

6. Understanding of Industrial Workflow and Standards

Understood the entire industrial workflow from designing the system, programming the system, wiring the system, testing, commissioning, and maintenance. Industrial safety and standard operating procedures were also stressed during the training.

7. Personal Growth and Confidence Building

My confidence in using industrial automation equipment for myself has grown after the practical exposure. This was also an inspiration to learn advanced technologies in PLC, HMI, and Industrial Control systems.

8. Overall Outcome

Overall, the industrial training was very effective in bridging the gap between the classroom and industry. It helped me build my technical knowledge, hands-on experience and professional disposition which I can apply to future academic projects and industrial automation.

9. SELF REFLECTION

The six-week industrial training at the Wonder Automation was a great learning opportunity that helped me broaden my understanding of industrial automation, PLC programming and HMI development. Prior to the training, what I knew about the concepts in Digital Electronics, Control Systems, related subjects were just those that were taught in my class. Doing the things in a real industrial environments, I was able to comprehend the practical application of these concepts.

In the first week, I learned about the PLC hardware, how to program the PLC using ladder logic, industrial sensors, relays, and basic wiring. It was not easy to get started with understanding PLC programming and industrial automation. But with constant practice and the assistance of trainers these ideas came to be familiar to me. This phase was very useful in my learning as it taught me something about precision, safety and detail.

As I continued the training, I became more confident in writing ladder logic programs, troubleshooting errors, creating HMI screens using Kinco's Builder software. The most interesting aspect of training was how to communicate between the PLC and the HMI. The development and solutions of programming and wiring errors enhanced my analytical thinking and problem solving.

The last stage of the training was practical, where I was able to put all this knowledge into practice by programming, wiring, testing and debugging. These tasks were completed successfully and helped me feel good about myself and my ability to work with industrial automation systems.

In addition to technical knowledge, I gained many other benefits from the training including communication skills, team work, time management and professional discipline. I learned the importance of Safety Practice, proper documentation and systematic working process to complete assigned tasks.

Overall, I have found it to be an excellent industrial training both technically and personally. It changed my perception of industrial automation from theory to practical, which I can use in actual situations. I am very interested in the PLC programming, HMI designing, and industrial control systems and I want to learn more in the future. I think that I will be able to use the knowledge and experience acquired during the training for my academic project, for my future internships and my professional career in the field of industrial automation.

10. CONCLUSION

The industrial training conducted by Wonder Automation Training Division, Mohali was able to achieve its target of giving hands-on experience to the students on industrial automation and PLC-based control system. The training lasted for 6 weeks, comprised of basic industrial automation, programming of PLC, industrial communication, sensors, actuators, VFDs and Industrial Control Applications. The programme was well designed and simultaneously theoretical and practical, giving a thorough and complete knowledge on the automation system in the modern industry.

The training was developed in a progressive manner from simple to complex. In the first few weeks, I was introduced to the concepts of automation, electrical control components, and PLC programming. During the next few weeks, I was able to get hands-on experience in industrial communication, networking, control devices and automation software. The last stage was all about configuring multiple automation elements and applying the learned knowledge to actual industrial automation applications.

One of the main components of the training involved completing a mini-project whereby I was able to put into practice all the principles learnt throughout the programme within a practical setting. It has helped me to become a better programmer, troubleshooter, problem solver, and to be more confident in dealing with industrial automation systems. It also emphasized the need for systematic planning, testing and appropriate documentation in the development of a project.

Industrial safety, industrial wiring rules, programming accuracy and followed engineering rules were consistently covered. They helped develop my reliability, accuracy, and discipline skills, which were important in industrial settings, and were a preparation for my professional duties.

The industrial training overall was successful, it helped in bridging the gap between academia and industry. It enhanced my technical know-how, developed my practical abilities and boosted my confidence when working with automation technologies. This experience has given me a good foundation for the future academic projects and career in industrial automation and control engineering..

11. FUTURE OUTLOOK AND SCOPE

1. Growing Demand for Industrial Automation

As the use of smart factory and digital production systems is expanding, the industrial automation is one of the fastest growing areas. The automotive, pharmaceuticals, food processing, power generation, oil and gas, textiles, and manufacturing industries are looking for employees who have an understanding of PLCs, industrial communication, sensors, and control systems. There is a growing need for automation engineers in industries that are looking for ways to increase productivity, cut down on operational costs, and improve product quality.

2. Advancement of Technology and Innovation

Industrial automation is a constantly changing landscape, incorporating technologies like Artificial Intelligence (AI), Industrial Internet of Things (IIoT), cloud computing, robotics, digital twins and predictive maintenance. These technologies allow real-time monitoring, intelligent decision making, remote diagnostics, and increased operational efficiencies. With the advent of Industry 4.0, automation experts need to be familiar with these new trends.

3. Opportunities for Higher Learning and Specialization

Advanced certifications and specialized courses are available for students interested in industrial automation, enhancing their technical skills. There are several good certifications to consider:

- Siemens Certified Programmer
- Rockwell Automation PLC Certification
- Schneider Electric EcoStruxure Training
- Mitsubishi PLC Programming Certification
- ISA Certified Control Systems Technician (CCST)
- Industrial Robotics and IIoT Certification Courses

4. Industrial Relevance and Career Prospects

After acquiring the knowledge in Industrial Automation, one can pursue the following career options:

- **Automation Engineer** – Designs, programs and maintains industrial automated systems.
- **PLC Programmer** – Designs and debugs PLC programs to control industrial processes.
- **Control Systems Engineer** – Designs and Optimises Control Systems for Manufacturing Plant.
- **Instrumentation Engineer** – Deals with sensors, transmitters in industry and measuring instruments.

- **Maintenance Engineer** – Ensures reliable operation of automated production equipment.
- **Industrial Automation Service Engineer** – Installs, commissions, and supports automation systems at industrial sites.

5. Contribution to Sustainable and Smart Development

Industrial automation is an important part of a sustainable and smart production. Automated systems optimise energy use, minimise material waste, drive down production errors and optimise resource use. They also improve job safety, minimize human contact with hazardous tasks and enable eco-friendly industrial processes.

6. Personal Career Vision

The training has been a motivating experience and has given me a deeper interest in the field of industrial automation, inspiring me to gain proficiency in PLC programming, industrial communication, and control systems. My goals for the next few years are to get more practical industrial experience, obtain professional certifications, remain an automation engineer, and continue to expand my knowledge and expertise through hands-on projects and lifelong learning.

7. Long-Term Industry Perspective

Smart factories, AI-powered manufacturing, collaborative robots, IIoT and data-informed decision-making will be the key to industrial automation of the future. Automation systems will get smarter, smarter and smarter. My working life will be committed to continuous skill development, acquisition of new technologies and adaptation to new industrial standards and innovations to stay relevant in this rapidly evolving field.

12. REFERENCES

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